

SADIA AFRIN

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SUMMARY

High-achieving EEE graduate from BRAC University (CGPA 3.71) with research interests in smart grids, AI-driven cybersecurity, and electric vehicle infrastructure. Co-author of six Q1 journal publications in smart grid stability, cybersecurity, and EV digital twins. Former Student Tutor, Student Mentor, and Vice President of the BRAC University Electrical and Electronic Club (BUEEC), with strong leadership, mentorship, and project management experience. Seeking to pursue an M.Sc. at a leading UK university.

EDUCATION

Bachelor of Science in Electrical and Electronic Engineering <i>BRAC University, Department of EEE, BSRM School of Engineering</i>	Jan 2022 – Jan 2026 CGPA: 3.71
Higher Secondary Certificate (HSC) <i>SOS Hermann Gmeiner College, Dhaka Board</i>	2020 GPA: 5.00
Secondary School Certificate (SSC) <i>SOS Hermann Gmeiner College, Dhaka Board</i>	2018 GPA: 5.00

PROFESSIONAL EXPERIENCE

Club Vice President <i>BRAC University Electrical and Electronic Club (BUEEC)</i>	Dec 2024 – Mar 2026
Coordinated club events and led cross-functional teams, strengthening strategic planning, team-building, public speaking, and negotiation skills.	
Student Tutor (ST), BSRM SoE, Dept. of EEE <i>BRAC University</i>	Fall 2024 – Summer 2025
Collaborated with peers and faculty members to support undergraduate learning, sharpening leadership, mentoring, adaptability, and empathy.	
Student Mentor, First Year Advising Team (FYAT) <i>BRAC University</i>	Summer 2024 – Fall 2025
Guided first-year mentees through academic and personal challenges, honing communication, conflict resolution, mentoring, and motivational skills.	

COURSES TAUGHT AS STUDENT TUTOR

EEE/ECE305: Control Systems <i>BRAC University</i>	Fall 2024 – Summer 2025
Supported a first course on feedback control of dynamic systems, covering modeling, analysis, and design of continuous-time linear feedback control systems, including time- and frequency-domain performance analysis, classical control design techniques for synthesizing linear controllers to meet stability and response criteria, case studies of real control systems, and MATLAB/Simulink for simulation and design.	

PROJECTS

IoT-Based Monitoring and Automated Air Purification System for Industrial Pollution Control <i>Final Year Design Project</i>	Jan 2025 – Jan 2026
- Designed an IoT-enabled monitoring system to track real-time air quality and pollution levels in industrial environments. - Developed an automated purification mechanism triggered by sensor thresholds to reduce airborne industrial pollutants. - Validated system performance through industry testing, confirming reliability under real operating conditions.	

Strategic Site Selection and Scheduling Optimization for Public Electric Vehicle Infrastructure

Design Project

Completed

- Developed an optimization framework for selecting strategic locations for public EV charging infrastructure.
- Modeled charging demand and scheduling constraints to improve infrastructure utilization and accessibility.
- Applied data-driven analysis to balance cost-efficiency with public accessibility for proposed EV charging sites.

PUBLICATIONS

1. **AI-powered cybersecurity for smart grid communication: A systematic review of intrusion detection and threat mitigation systems**
Authors: Sadia Afrin, Md. Reshad Al Muttaki, Alvi Ibn Amzad Anil, Shameem Hasan
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **1st Author**
DOI: [10.1016/j.ecmx.2025.101416](https://doi.org/10.1016/j.ecmx.2025.101416)
2. **Towards smarter grids: A systematic review of wide area monitoring enhancing stability, protection, and promoting environmental sustainability**
Authors: Md. Reshad Al Muttaki, Alvi Ibn Amzad Anil, Sadia Afrin, Shameem Hasan
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **1st Author**
DOI: [10.1016/j.ecmx.2025.101424](https://doi.org/10.1016/j.ecmx.2025.101424)
3. **State-of-charge-aware charging opportunity detection for electric vehicles using data-driven learning and digital twin simulation**
Authors: Md. Reshad Al Muttaki, Sadia Afrin, Alvi Ibn Amzad Anil, Shameem Hasan
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **2nd Author**
DOI: [10.1016/j.ecmx.2026.101775](https://doi.org/10.1016/j.ecmx.2026.101775)
4. **The next generation of energy storage for smart and sustainable power grids: A systematic review**
Authors: Alvi Ibn Amzad Anil, Md. Reshad Al Muttaki, Sadia Afrin, Shameem Hasan
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **3rd Author**
DOI: [10.1016/j.ecmx.2026.101771](https://doi.org/10.1016/j.ecmx.2026.101771)
5. **Demand-based anomaly detection algorithm in electric vehicle charging systems using machine learning and digital twin simulation**
Authors: Md. Reshad Al Muttaki, Sadia Afrin, Alvi Ibn Amzad Anil, Shameem Hasan
Journal of Energy Storage | Q1, Top 4% | IF: 9.8 | CiteScore: 13.3 | **2nd Author**
DOI: [10.1016/j.est.2026.122795](https://doi.org/10.1016/j.est.2026.122795)
6. **Balancing cybersecurity, wireless charging, and safety: Digital twin-driven techno-economic analysis of electric vehicles**
Authors: Shameem Hasan, Abu M. Fuad, B.M. Khalid Hasan Drobo, Md. Reshad Al Muttaki, Sadia Afrin, Alvi Ibn Amzad Anil, Omar Farrok
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **5th Author**
DOI: [10.1016/j.ecmx.2026.102107](https://doi.org/10.1016/j.ecmx.2026.102107)

RESEARCH INTERESTS

Electronics & Semiconductor: VLSI Design, Semiconductor Devices, Analog & Digital Circuit Design, Embedded Systems

Smart Grid & Energy Systems: Smart Grid Communication, Cybersecurity for Power Systems, Electric Vehicle Infrastructure, Digital Twin Systems

Artificial Intelligence & Machine Learning: AI-Driven Cybersecurity, Machine Learning for Predictive Analytics, Anomaly Detection, Data-Driven Optimization, AI & ML for Education

Achievements and Honors

- **Vice Chancellor's (VC's) List** — Summer 2024
- **Dean's List** — Spring 2023; Summer 2023, Fall 2023; Fall 2024; Spring 2025

SKILLS

Programming: Python, C, MATLAB, Verilog, VHDL

EDA & Simulation: Cadence, Altium Designer

Tools & Platforms: Google Suite, MS Office Suite

Core Competencies: Project Management, Leadership, Mentorship, Critical Thinking, Creativity

LANGUAGE PROFICIENCY

English – IELTS

Overall Band Score: 7.0

Listening: 7.5

Reading: 7.0

Writing: 6.0

Speaking: 8.0

REFERENCES

Dr. Mirza Rasheduzzaman

Associate Professor

Department of EEE, BRAC University

mirza.rasheduzzaman@bracu.ac.bd

Shameem Hasan

Senior Lecturer

Department of EEE, BRAC University

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